

Sprint Report 4

PenTestPocket

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November 9th 2025 - December 7th 2025

Section 1: What's New (User Facing)

1.1 Feature - Pinned Tools with Saving

In the tool pinning section on the main page, if you select the pin icon on the right, four drop downs appear with the option to select a tool. Selecting the pin icon again allows you to go to that tools page. And with the help of the JSON file, if you close PTP and relaunch your pins will stay saved

1.2 Feature - Interactive Built in Terminal*

Clicking on the right most icon it opens a built terminal for PTP that acts like the built in system terminal, with the exception that at this stage the terminal is not functioning at this stage, but has the ability to be working with few additions

1.3 Feature - Enhanced User Logging

Every action done by the user is tracked and store within the userlogs.txt and is displayable inside of PTP

Section 2: Work Summary (Developer Facing)

In **Sprint 4**, I streamlined the code to not have print statements for debugging, and removed conditional checkers that became obsolete as the project advanced. Although not all backend pieces were completed, the tool has functioning capabilities to show as a proof of concept

This was all accomplished starting from August 21st 2025, time tracking is stored in GitHub to break down how much time was spent in each grouping (specified in the roadmap) and how much each day.

Section 3: Unfinished Work

In **Sprint 4** I was not able to finish all tools originally planned, no automation, no internal notebook, and not full in-depth tools. The default reason, which is true for these issues as well, is that for any delayed issue it is due to the tasks taking longer than expected; any other arising issues will be mentioned within the comments of the issue itself. For this sprint, a lot of delays came in the form of goal changes, other coursework, and personal commitments.

NOTE - There are open issues, but they are meant to be placed into the final sprint, so they are not in scope for this sprint.

Section 4: Completed Issues/User Stories

Here are links to the issues that I completed in this sprint:

4.1 Issue links

Issue #10 - Project Coaching

Issue #21 - Weekly Coaching Session - 11/06

Issue #22 - Weekly Coaching Session - 11/13

Issue #23 - Poster Session - 12/4

Issue #24 - Weekly Coaching Session - 11/20

Issue #33 - Poster Abstract Report Due

Issue #41 - Upload Time Tracker

Issue #55 - Create more in depth user logging

Section 5: Incomplete Issues/User Stories

Here are links to issues I worked on but did not complete in this sprint:

5.1 Issue links & Explanations

[Will be Completed] Issue #32 - Sprint Four Report Due

Explanation: This is the Sprint Report Four, if you are seeing this, it is completed

[Will be Completed] Issue #37 - Project Draft Four Report Due

Explanation: Project Four Report will be submitted before December 7th (end of Sprint 4 and Capstone)

[Incomplete] Issue #43 - Make all elements on main screen scale with the screen

Explanation: End of Capstone term, out of time

[Incomplete] Issue #46 - Add authorization

Explanation: End of Capstone term, out of time

[Incomplete] Issue #53 - Assign Ncat Commands

Explanation: End of Capstone term, out of time

[Incomplete] Issue #56 - Store tool output in a file for later

Explanation: End of Capstone term, out of time

[Incomplete] Issue #58 - Integrate Metasploit

Explanation: End of Capstone term, out of time

[Incomplete] Issue #59 - Assign metasploit commands

Explanation: End of Capstone term, out of time

[Incomplete] Issue #61 - Assign curl commands

Explanation: End of Capstone term, out of time

[Incomplete] Issue #62 - Integrate gobuster

Explanation: End of Capstone term, out of time

[Incomplete] Issue #63 - Assign gobuster commands

Explanation: End of Capstone term, out of time

[Incomplete] Issue #66 - Assign hashcat commands

Explanation: End of Capstone term, out of time

Section 6: Code Files for Review

Please review the following code files, which were actively developed during this sprint, for quality:

Main PenTestPocket Python Code

Change Log Python Code

User Logging Python Code

Help Window Python Code

Service Scanning Python Code

Password Cracking Python Code

Web Enumeration Python Code

Encode/Decode Python Code

Create Hash Python Code

Section 7: Retrospective Summary

7.1 What went well

Implementation of tools - All tools that were implemented are implemented to a level of clarity of speed such that they are no sacrifices between other tools

Code structure - The structure of the code is designed in a way that is easily readable, grouped together in sections, and decoupled in a powerful architectural design

Scalability - With the architectural design, PTP is built in a way such that any and all additions would not require any refactoring or redesign

7.2 What needs improvement

Storing run execution data - Tool output execution isn't saved anywhere after the tool is run, it is only run and then displayed and once closed it is gone forever

Help Window - The help window can be opened but it is completely blank, the help text needs to be created and added to the help window

Change Logs - Similar to the help window, the change log window can be opened but no text is present, but the text that would be included is in readME.md

7.3 Future Work

Breadth of Tools - The amount of tools is not the total I aimed when I created PTP, so the remainder of the unimplemented tools need to be added to get the full breadth of PTP

Cross-Platform enabled - PTP currently only is tested working on debian linux 12, ideally I would like it to run on other distros of linux alongside with Windows as it is the most popular operating system

Ensuring constant compliance with the CIA triad -

Currently I am not doing anything to ensure CIA, but for each aspect I would do the following. Confidentiality, I would add user authentication and 2FA to ensure only authorized users can access information. Integrity, I would take hashes of PTP when changes occur, and on relaunch compare hashing to see if unauthorized changes were made. Lastly for availability, this is the only aspect that isn't really needing much change, the only change I would make is if bugs occur the program fails safely instead of completely shutting down